XRPL Validator — Daily Engineering Report · 2026-10-27

Period: 2026-10-27 00:00 to 24:00 US Eastern (2026-10-27 04:00 to 2026-10-28 04:00 UTC); ledgers #107,929,025 to #107,950,579. Every count from the ledger and the validator's logs stops at #107,950,579, closed 03:59:52 UTC. Generated 2026-10-28 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,555 of the 21,555 ledgers the network closed (100.000%); every validation matched the network's validated ledger.
- **Native engine:** 0 engine receipts.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Independence program:** 0 of 30 qualifying days (snapshot at 04:20:35 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,555
Validations issued	21,555
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	21,555 of 21,555
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	0
Basis: our own ledger hash	21,548
Basis: our own hash, witnessed by the reference node (quorum short)	7
External agreement, data.xrpl.org (UTC day 2026-10-27)	1.00000 (0 missed of 20,691)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,929,025–#107,950,579), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 04:20:35 UTC (since the last restart, 2026-10-25 21:53 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	48,601 ledgers, 9,460,155 transactions	48,601 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	48,932 ledgers, 9,707,486 records	9,707,484 exact; differ 2, tree 2
3 · transaction order	48,932 ledgers (mode: on)	48,932 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	48,953 ledgers fetched	48,953 matched; differ 0, failed 0
4 · trusted validations	49,004 ledgers judged	49,004 agreed; differ 0, short of quorum 0
5 · our own ledger hash	49,006 ledgers	48,928 own; differ 2, partial 76
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **0 of 30** (snapshot at 04:20:35 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-27 — tracked #107,925,424–#107,947,015	Value
Validations	21,592 (21,592 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	no

Not qualifying: the ledger data came from the reference node, not peers (Step 4).

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,555 (#107,929,025–#107,950,579)
Mean close interval	4.01 s
Longest close interval	10 s (before #107,950,553)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Committed, awaiting merge and deployment: `b812aff` docs(step4): the peer switch plan — ledgers from peers, a non-signing twin to prove it.
- Committed, awaiting merge and deployment: `d6f8f98` feat(step4): XRPL_LEDGER_SOURCE switch; peer mode refuses and counts every RPC call.
- Committed, awaiting merge and deployment: `ce74ac9` feat(step4): own_validated says when a newer ledger reaches quorum.
- Committed, awaiting merge and deployment: `d1050b4` feat(step4): peer_fetch serves awaited ledger, header and state jobs; keyed state fetch.
- Committed, awaiting merge and deployment: `91e44f3` feat(step4): peer_ledgers — the validated hash for any seq and its inputs from peers.
- Committed, awaiting merge and deployment: `f81c4ab` feat(step4): the native writer hands over its own system-object writes.
- Committed, awaiting merge and deployment: `4e3b50e` feat(step4): the validator's ledger feed takes each ledger from peers when XRPL_LEDGER_SOURCE=peers.
- Committed, awaiting merge and deployment: `491ed2b` feat(step4): warm restart from peers; the last the reference node reads go.
- Committed, awaiting merge and deployment: `3b8e57f` test(step4): the quorum wait tests run on real time (no tokio test-util).
- Committed, awaiting merge and deployment: `c40ae64` feat(step4): XRPL_WITNESS_RPC and XRPL_OBSERVE; /api/engine shows the source.
- Committed, awaiting merge and deployment: `54b7b01` test(step4): the observe test installs rustls's provider as live_viewer does.
- Committed, awaiting merge and deployment: `7ac4419` feat(step8): cold start from peers — proven leaves become a store live_viewer resumes.
- Committed, awaiting merge and deployment: `52396bd` fix(paths): every receipt and state file the validator writes follows XRPL_DATA_DIR.
- Committed, awaiting merge and deployment: `6b6e5b6` feat(step4): the twin — a peer-fed, non-signing copy and its check.
- Committed, awaiting merge and deployment: `5636f20` fix(step4): the build without ffi compiles its tests again; one stream closure becomes a fn.
- Committed, awaiting merge and deployment: `ba1f8cf` fix(step4): review findings — no FFI repair RPC from peers, a warm stop never waits on a peer fetch, peers need the writer.
- Committed, awaiting merge and deployment: `fb518ce` docs(step4): the final review's findings and fixes.
- Committed, awaiting merge and deployment: `ea798bf` fix(step8): the peer cold start takes the live launcher's list pin and runs at the lowest priority.
- Committed, awaiting merge and deployment: `fd7ca57` fix(step4): from peers the mirror loads at once, however far behind.

- Committed, awaiting merge and deployment: `7ed1408` feat(5a): XRPL_OWN_BREAK — the breaker's wait, or off for the 30-day count.
- Committed, awaiting merge and deployment: `3af6c1a` feat(step4): twin-watch.sh — the twin check once an hour, to the twin's data dir.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.